

Tableau Daily KPI Dataset Export

Online Retail II Data Pipeline Project

Overview

After building the analytics warehouse, I created a separate dataset specifically designed for visualization in Tableau. Since Tableau Public does not support direct connections to PostgreSQL databases, the dashboard needs to read from a file-based dataset.

To solve this, I wrote a Python script that exports a curated dataset from the analytics layer into a CSV file that Tableau can read directly.

The script responsible for this export is:

`pipeline_scripts/11_export_tableau_daily_kpis.py`

This step allows the pipeline to automatically generate a Tableau-ready dataset whenever the analytics layer is refreshed.

Export Process

The export script performs several steps.

First, it connects to the PostgreSQL database using environment variables defined in the `.env` file. These variables include the database host, port, database name, user, and password. After establishing the connection, the script queries the analytics warehouse using the following tables:

- `analytics.fact_sales`
- `analytics.dim_date`
- `analytics.dim_country`

The query aggregates data at the daily and country levels, producing a dataset that is small enough for efficient dashboard loading while still supporting useful analysis.

Metrics Generated

The exported dataset includes several key performance metrics:

- `daily_orders` — number of distinct invoices per day
- `gross_revenue` — total positive revenue values
- `returns_revenue` — revenue associated with returned items
- `net_revenue` — total revenue including returns
- `is_q4` — indicator identifying whether a date falls in the fourth quarter

These fields provide the core metrics required to build a retail performance dashboard.

Data Types and Schema Handling

The Python script does not explicitly cast data types when exporting the dataset. Instead, the PostgreSQL data types are preserved when the query results are loaded into pandas.

As a result, numeric and date fields remain correctly typed when the dataset is exported. Tableau then interprets these values automatically when the CSV file is loaded.

The only derived field created during the export process is the Q4 indicator, which is generated directly in the SQL query.

Why the Data Is Aggregated

The exported dataset is intentionally aggregated instead of including the full transactional table. The original fact table contains more than one million rows. Exporting the entire dataset would make dashboards load more slowly and be harder to maintain.

Instead, the export step produces an aggregated dataset containing 3,207 rows of daily observations. This smaller dataset is easier for visualization tools to process while still preserving the key metrics required for analysis.

This approach follows a common data engineering practice where a serving layer is created for business intelligence tools.

In this architecture:

- The data warehouse stores detailed transactional data
 - The serving layer provides curated, aggregated datasets optimized for reporting and dashboards
-

Outcome

By exporting an aggregated dataset specifically designed for visualization, the pipeline separates the analytics warehouse from the dashboard layer.

This makes the system easier to maintain and improves dashboard performance. Tableau can load the dataset quickly, while the warehouse continues to store the full transactional history for deeper analysis when needed.